

Exercises: Multiple Linear Regression Part II

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(Referring to the theoretical parts: 9, 10, 12, 13, 14, 15, 16, 17, 18, 19)

1 Exercise 1

Among $n = 31$ cherry trees, data about *Volume*, *Diameter* and *Height* of each tree were collected.

Exercise 1.1

Specify an appropriate regression model for the response variable *Volume* and the related assumptions. Propose a transformation for all variables by thinking about the geometric relationship of our variables.

$$n = 31 \quad \begin{array}{l} \cdot \text{VOLUME} \rightarrow V \\ \cdot \text{DIAMETER} \rightarrow d \\ \cdot \text{HEIGHT} \rightarrow h \end{array} \quad \begin{array}{c} \text{Cylinder} \\ \sim \end{array} \quad V = \pi r^2 h = \frac{\pi}{4} d^2 h$$

- $X_{:,1} = x_{:,1} ; X_{:,p} = x_{:,p}$
- $E(\epsilon_i) = 0 \quad V_i : i = 1, \dots, 31$
- $\text{VAR}(\epsilon_i) = \sigma^2 \mathbb{I}$
- $\text{COV}(\epsilon_i; \epsilon_k) = 0 \text{ for } i \neq k \rightarrow \text{COV}(Y_i; Y_k) = 0 \text{ for } i \neq k$
- $\epsilon_i \stackrel{\text{ID}}{\sim} \mathcal{N}(0; \sigma^2)$
- VECTORS $x_j \in \mathbb{R}^n$ ARE LINEARLY INDEP. $\rightarrow \text{RANK}(X) = P$ (FULL RANK)
 $J = 1, \dots, P$

$$V = \frac{\pi}{4} d^2 h \rightarrow \log(V) = \log\left(\frac{\pi}{4}\right) + 2\log(d) + \log(h)$$

$$\tilde{V} = \log\left(\frac{\pi}{4}\right) + 2\tilde{d} + \tilde{h}$$

$$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}$$

$$\beta_1 = \log\left(\frac{\pi}{4}\right) \quad Y_i = \tilde{V}_i = \log(V)$$

$$\beta_2 = 2 \quad x_{1i} = \tilde{d} = \log(d)$$

$$\beta_3 = 1 \quad x_{2i} = \tilde{h} = \log(h)$$

Exercise 1.2

Knowing that

$$(X^T X)^{-1} = \begin{bmatrix} 96.572 & 3.139 & -24.165 \\ 3.139 & 0.849 & -1.227 \\ -24.165 & -1.227 & 6.310 \end{bmatrix} \quad X^T \mathbf{y} = \begin{bmatrix} 101.455 \\ 263.056 \\ 439.896 \end{bmatrix}$$

Find the estimate for β and write the estimated model.

$$\hat{\beta} = (X^T X)^{-1} X^T y = \begin{bmatrix} -6.648 \\ 2.049 \\ 1.3139 \end{bmatrix} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix}$$

(3x3)(3x1) = (3x1)

$$\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_{1i} + \hat{\beta}_3 x_{2i}$$

↓

$$\hat{y}_i = -6.648 + 2.049 \cdot x_{1i} + 1.3139 \cdot x_{2i}$$

$$\mathbf{y} = \mathbf{X} \beta + \epsilon \rightarrow \begin{matrix} \mathbf{y} \\ (13 \times 1) \end{matrix} = \begin{matrix} \mathbf{X} \\ (13 \times 3) \end{matrix} \begin{matrix} \beta \\ (3 \times 1) \end{matrix} + \begin{matrix} \epsilon \\ (13 \times 1) \end{matrix} \quad \epsilon \sim N(0, \sigma^2 I)$$

Exercise 1.3

Knowing the maximum likelihood estimate $\hat{\sigma}^2 = 0.00598$, find the unbiased estimate $\hat{s}^2$.

Then, calculate the estimated variance-covariance matrix of $\hat{\beta}$.

$$\hat{\sigma}^2 = 0.00598 \xrightarrow{\text{UNBIASED ESTIMATE}} \hat{s}^2 = \frac{n}{n-p} \hat{\sigma}^2 = \frac{31}{31-3} \cdot 0.00598 = 2.135 \cdot 10^{-4} = 0.0006207 = 6.6 \cdot 10^{-5}$$

VARIANCE-COVARIANCE MATRIX

$$\begin{bmatrix} \text{VAR}(\hat{\beta}_1) & \text{COV}(\hat{\beta}_2; \hat{\beta}_1) & \text{COV}(\hat{\beta}_3; \hat{\beta}_1) \\ \text{COV}(\hat{\beta}_1; \hat{\beta}_2) & \text{VAR}(\hat{\beta}_2) & \text{COV}(\hat{\beta}_3; \hat{\beta}_2) \\ \text{COV}(\hat{\beta}_1; \hat{\beta}_3) & \text{COV}(\hat{\beta}_2; \hat{\beta}_3) & \text{VAR}(\hat{\beta}_3) \end{bmatrix} = \begin{bmatrix} 0.639 & 0.0208 & -0.16 \\ 0.0208 & 0.0056 & -0.0081 \\ -0.16 & -0.0081 & 0.042 \end{bmatrix}$$

- $\text{VAR}(\hat{\beta}_1) = s^2 \cdot c_{11} = 2.135 \cdot 10^{-4} \cdot 96.572 = 6.6 \cdot 10^{-5}$
- $\text{VAR}(\hat{\beta}_2) = s^2 \cdot c_{22} = 2.135 \cdot 10^{-4} \cdot 0.249 = 6.6 \cdot 10^{-5}$
- $\text{VAR}(\hat{\beta}_3) = s^2 \cdot c_{33} = 2.135 \cdot 10^{-4} \cdot 6.310 = 6.6 \cdot 10^{-5}$
- $\text{COV}(\hat{\beta}_1; \hat{\beta}_2) = \text{COV}(\hat{\beta}_2; \hat{\beta}_1) = s^2 \cdot c_{12} = 2.135 \cdot 10^{-4} \cdot 3.139 = 6.6 \cdot 10^{-5}$
- $\text{COV}(\hat{\beta}_1; \hat{\beta}_3) = \text{COV}(\hat{\beta}_3; \hat{\beta}_1) = s^2 \cdot c_{13} = 2.135 \cdot 10^{-4} \cdot (-24.165) = -6.6 \cdot 10^{-5}$
- $\text{COV}(\hat{\beta}_2; \hat{\beta}_3) = \text{COV}(\hat{\beta}_3; \hat{\beta}_2) = s^2 \cdot c_{32} = 2.135 \cdot 10^{-4} \cdot (-1.127) = -6.6 \cdot 10^{-5}$

$$\hat{\beta} \sim N_p(\beta; \sigma^2 (X^T X)^{-1})$$

AND EACH ESTIMATOR
 $\hookrightarrow \hat{\beta}_i \sim N(\beta_i; \sigma^2 (X^T X)^{-1}_{ii})$

Exercise 1.4

Perform a statistical test to evaluate if the regression coefficient related to the *Diameter* is significant. Specify the system of hypothesis, the test statistic and the p-value. Then, perform another statistical test with the following null hypothesis $H_0 : \beta_2 = \beta_3 = 0$ (Restricted model SSE = 8.309).

$$\text{Mod}_A: \hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_{1i} + \hat{\beta}_3 x_{2i}$$

$$\text{Mod}_B: \hat{y}_i = \hat{\beta}_1 + \hat{\beta}_3 x_{2i}$$

$$SSE = \frac{S^2}{h-p}$$

$$S^2 = \frac{SSE}{h-p} \rightarrow SSE_A = 0.0066207 \cdot (31-3) = 0.1853$$

$$S^2 = 0.0066207$$

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

$$f_{obs} \stackrel{H_0}{\sim} f_{h-p; p-p} \quad f_{obs} = \frac{\frac{SSE_A - SSE_B}{p-p}}{\frac{SSE_A}{h-p}} \quad \text{NOT DOUBLE SINCE } SSE_B \text{ IS UNKNOWN}$$

$$t_2^{obs} \stackrel{H_0}{\sim} t_{h-p} \quad t_2^{obs} = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} = \frac{2.049 - 0}{\sqrt{0.0056}} = 27.386$$

↓
VAR($\hat{\beta}_2$)

$$\alpha^{obs} = P(K_{h-p} \geq |t_2^{obs}|) = 2 \min \{ P(t_{h-p} < -t_2^{obs}), P(t_{h-p} > t_2^{obs}) \} \sim 2(1 - \dots) \sim 0$$

$h-p = 31-3 = 28$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1$;
 $\alpha = 0.05$; $\alpha = 0.01$ WE REJECT
 H_0 AT ALL CONFIDENCE LEVEL

↓
WE PREFER MODEL WITH DIAMETER
THAN THE MODEL WITHOUT DIAMETER

$$\text{Mod}_A: \hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_{1i} + \hat{\beta}_3 x_{2i}$$

$$\text{Mod}_C: \hat{y}_i = \hat{\beta}_1$$

$$SSE = \frac{S^2}{h-p}$$

$$S^2 = \frac{SSE}{h-p} \rightarrow SSE_A = 0.0066207 \cdot (31-3) = 0.1853$$

$$SSE_C = 8.309$$

$$p=1$$

$$S^2 = 0.0066207$$

$$p=3$$

$$f_{obs} \stackrel{H_0}{\sim} f_{p-p_0; h-p} \quad f_{obs} = \frac{\frac{SSE_C - SSE_A}{p-p_0}}{\frac{SSE_A}{h-p}} = \frac{\frac{8.309 - 0.1853}{3-1}}{\frac{0.1853}{31-3}} = 413$$

$$\alpha^{obs} = P(f_{p-p_0; h-p} > f_{obs}) \sim 1 - \dots \sim 0$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$
WE REJECT H_0 AT ALL CONFIDENCE LEVEL

↓
WE PREFER THE MODEL WITH BOTH HEIGHT AND
DIAMETER WITH RESPECT TO THE MODEL
WITH ONLY THE INTERCEPT

2 Exercise 2

Among n=32 births, the following three variables were collected:

- *Weight*: birth weight in grams of baby
- *Smoking*: Smoking status of mother (yes or no)
- *Gest*: length of gestation in weeks

Describe the equation of the multiple regression model specifying the nature of each variable (without including the interaction term). Then, express the model in matrix form: $\underline{Y} = X\beta + \underline{\varepsilon}$, explicitly stating $\underline{Y}$, X , β , and $\underline{\varepsilon}$ (and its distribution).

$n = 32$
 • WEIGHT
 • DIMENSIONS
 • COST

$\vec{y} = \beta_0 + \beta_1 D_1 + \beta_2 X_2 + \epsilon$
 (Annotations: β_0 is WEIGHT, β_1 is DIMENSIONS, β_2 is COST, D_1 is 1 Mother Yes / 0 Mother No)

$\vec{y} = X\beta + \epsilon$

Matrix dimensions: $(32 \times 1) = (32 \times 3) + (32 \times 1) + (32 \times 1)$

$\epsilon \sim N(0; \sigma^2 I)$
 $\vec{y} \sim N(\vec{X}\beta; \sigma^2 I)$

• $E(\epsilon_i) = 0$ for $i = 1, \dots, 32$
 • $\text{VAR}(\epsilon_i) = \sigma^2 > 0$ for $i = 1, \dots, 32$
 • $\text{Cov}(\epsilon_i; \epsilon_k) = 0$ for $i \neq k \rightarrow \text{Cov}(\vec{y}_1; \vec{y}_n) = 0$
 • $\epsilon_i \sim N(0, \sigma^2)$
 • $\mathcal{X}_1 = x_{1,1}; \mathcal{X}_2 = x_{1,2}$
 • THE VECTORS $\vec{x}_i \in \mathbb{R}^N$ ARE LINEARLY INDEP $\rightarrow \text{RANK}(X) = p$

Exercise 2.1

Knowing the following values

$$\widehat{Weight}^{(n)} = -2390 + 143 \text{ Gest} \quad (\text{Smoking} = 0)$$

$$\widehat{Weight}^{(y)} = -2635 + 143 \text{ Gest} \quad (\text{Smoking} = 1)$$

What is the estimated value of the regression coefficient associated to the dummy variable? Interpret the results and explain, theoretically, why the two equations are different.

$$\hat{Y}_i = \beta_0 + \beta_1 D_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

↑ WEIGHT ↑ SMOKING { 1 Mother yes
0 Mother no }
↑ GEST

$$D_{1i} = 0 \longrightarrow \hat{Y}_i = \beta_0 + \beta_2 X_{2i}$$

$$D_{1i} = 1 \longrightarrow \hat{Y}_i = (\beta_0 + \beta_1) + \beta_2 X_{2i}$$

$$\hat{\beta}_2 = 143$$

$$\hat{\beta}_0 + \hat{\beta}_1 = -2635 \rightarrow \hat{\beta}_1 = -245$$

$$\hat{\beta}_0 = -2390$$

- WHEN WE INCREASE THE LENGTH OF GESTATION OF ONE WEEK WE INCREASE THE WEIGHT OF THE BABY OF 143 GRAMS
- WHEN KEEPING GESTATION FIXED, IF WE MOVE FROM SMOKER TO NON-SMOKER ON AVERAGE WE INCREASE THE WEIGHT OF THE BABY OF 245 GRAMS

Exercise 2.2

After adding an interaction term, we obtained the following estimated regression model

$$\widehat{Weight} = -2546.138 + 147.207 \text{ Gest} + 71.574 \text{ Smoke} - 8.178 \text{ Gest} \times \text{Smoke}$$

Provide the estimated regression model for each group (Smoke: yes or no) and interpret the results.

SMOKING YES

$$\hat{Y}_i = (-2546.138 + 71.574) + (147.207 - 8.178) x_{2i}$$

SMOKING NO

$$\hat{Y}_i = (-2546.138) + 147.207 x_{2i}$$

- IF THE MOTHER IS A SMOKER IF WE INCREASE GESTATION OF ONE WEEK, WE INCREASE THE BABY WEIGHT OF 139.029 GRAMS ON AVERAGE
- IF THE MOTHER IS NOT A SMOKER, IF WE INCREASE GESTATION OF ONE WEEK, WE INCREASE THE BABY WEIGHT OF 147.207 GRAMS ON AVERAGE

THE INTERCEPT INTERPRETATION DOES NOT PROVIDE RELEVANT INFORMATION SINCE IT DOES NOT MAKE SENSE CONSIDER GEST :D

Exercise 2.3

Consider the following table where for the two models $\mathcal{M}_0$ (the restricted model, i.e. the model which just involves $Gest$) and $\mathcal{M}_4$ (corresponds to the model specified in the ex. 2.2) are expressed the residual sum of squares (SSE). Complete the table.

Model	D.o.f. $h-p$	SSE
Restricted ($\mathcal{M}_0$)	30	839951.03
Unconstrained ($\mathcal{M}_4$)	28	384391.46

Further, perform a statistical test with the following system of hypothesis (where Smoking has two classes: Y and N)

$$\begin{cases} H_0: \mu_N = \mu_Y \\ H_1: \mu_N \neq \mu_Y \end{cases}$$

Compute the test statistic and the p-value.

$$\begin{aligned} \text{Model 0: } Y_i &= \beta_0 + \beta_1 x_{i1}; & \text{Model 4: } Y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} \\ \text{SSE}_0 &= 839951.03 & \text{SSE}_4 &= 384391.46 \\ p &= 2 & p &= 4 \end{aligned}$$

$$\begin{aligned} &\begin{cases} H_0: \mu_N - \mu_Y = 0 \\ H_1: \mu_N - \mu_Y \neq 0 \end{cases} \\ f_{obs} &\overset{H_0}{\sim} f_{p-p_0; h-p} & f_{obs} &= \frac{\frac{SSE_0 - SSE_4}{p-p_0}}{\frac{SSE_4}{h-p}} = \frac{\frac{839951.03 - 384391.46}{4-2}}{\frac{384391.46}{31-4}} = 16.59 \end{aligned}$$

$$\alpha^{obs} = P(f_{p-p_0; h-p} > f_{obs}) \approx 1 - \sim \approx 0$$

Since $\alpha^{obs} < \alpha$ For $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ I REJECT
H₀ AT ALL CONFIDENCE LEVEL

Exercise 2.4

Now, we want just to test the regression coefficient related to the interaction. The SSE of the model without the interaction is equal to 387069.83. Perform a valid test and discuss about the results.

$$\text{Mod}_4: Y_i = \beta_0 + \beta_1 D_1 + \beta_2 x_2 + \beta_3 x_2 D_1$$

$$\text{SSE}_4: 384391.46$$

$$p=4$$

$$\text{Mod}_5: Y_i = \beta_0 + \beta_1 D_1 + \beta_2 x_2$$

$$\text{SSE}_5: 387069.83$$

$$p_0=3$$

$$\begin{cases} H_0: \beta_4 = 0 \\ H_1: \beta_4 \neq 0 \end{cases}$$

$$F_{obs} \stackrel{H_0}{\sim} F_{p-p_0; n-p} \quad F_{obs} = \frac{\frac{\text{SSE}_5 - \text{SSE}_4}{p-p_0}}{\frac{\text{SSE}_4}{n-p}} = 0.123$$

$$\alpha^{obs} = P(F_{1,27} > F_{obs}) \approx 1 - \dots \approx 0.6621$$

Since $\alpha^{obs} > \alpha$ for $\alpha=0.1$; $\alpha=0.05$; $\alpha=0.01$
WE CAN NOT REJECT H_0

Exercise 2.5

Let $SSE = 3735789.2$ be the Residual Sum of Squares of the model $Y_i = \beta_1 + \epsilon_i$. Compute R^2 ~~and explain the difference between R^2 and R^2 related to Y_i~~ . Explain the difference.

$$Y_i = \beta_1 + \epsilon_i$$

$$SSE = 3735789.2$$

$$SST = SSE \leftarrow \text{FOR MODEL WITH ONLY THE INTERCEPT}$$

$$R^2 = \frac{SST - SSE}{SST} = \frac{3735789.2 - 384391.46}{3735789.2} = 0.897$$

3 Exercise 3

Among $n = 15$ poisoned lab rats, data about survival time (Y) and antidote/treatment (T) were collected. The researchers consider three kind of treatment: A, B and C, where

Group	n_j NUMBER OF RATS IN EACH GROUP	$\bar{y}_j$ MEAN OF GROUP	s_j STANDARD DEVIATION
A ($j = 1$)	4	0.347	0.2055
B ($j = 2$)	3	3.067	0.4618
C ($j = 3$)	8	1.764	0.5007

Compute the total deviance (total sum of squares - SST) and explain its components.

$n = 15$

Group	n_j	$\bar{y}_j$	s_j
A ($j = 1$)	4	0.347	0.2055
B ($j = 2$)	3	3.067	0.4618
C ($j = 3$)	8	1.764	0.5007

$$\bullet \quad \bar{y} = \frac{1}{n} \sum n_j \bar{y}_j = \frac{1}{15} (4 \cdot 0.347 + 3 \cdot 3.067 + 8 \cdot 1.764) = 1.646$$

$$\bullet \quad SSE = \sum (n_j - 1) s_j^2 = (4 - 1)(0.2055)^2 + (3 - 1)(0.4618)^2 + (8 - 1)(0.5007)^2 = 2.3081$$

$$\bullet \quad SSR = \sum n_j (\bar{y}_j - \bar{y})^2 = 4(0.347 - 1.646)^2 + 3(3.067 - 1.646)^2 + 8(1.764 - 1.646)^2 = 12.918$$

$$SST = SSE + SSR = 15.21687$$

$$SST = \sum \sum (y_{ij} - \bar{y})^2$$

Exercise 3.1

Perform a statistical test to evaluate if the means (of each group) are homogeneous.
Specify the system of hypothesis, the test statistic and the p-value.

$$\begin{cases} H_0: \mu_1 = \mu_2 = \mu_3 \\ H_1: \neg H_0 \end{cases}$$

$$f_{0.95}^{H_0} \sim f_{3-1; 12-3} \quad f_{0.95} = \frac{\frac{SSR}{3-1}}{\frac{SSE}{12-3}} = \frac{\frac{12.92}{3-1}}{\frac{2.302}{12-3}} = 33.59$$

$$\alpha^{OBS} = P(F_{2,12} > 33.59) \approx 0 \quad \text{Since } \alpha^{OBS} < \alpha \text{ for } \alpha = 0.1; \alpha = 0.05, \alpha = 0.01 \text{ we} \\ \text{REJECT } H_0 \text{ AT ALL CONFIDENCE LEVEL}$$

Exercise 3.2

Specify an appropriate linear regression model. Given the previous data, find the estimates for each coefficient and interpret them.

$$Y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i$$

$$\begin{cases} Y_i \sim N(\beta_1, \sigma^2) & i=1, \dots, 4 \text{ GROUP A} \\ Y_i \sim N(\beta_1 + \beta_2, \sigma^2) & i=5, \dots, 7 \text{ GROUP B} \\ Y_i \sim N(\beta_1 + \beta_3, \sigma^2) & i=8, \dots, 10 \text{ GROUP C} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{IF ANTIDOTE B} \\ 0 & \text{OTHERWISE} \end{cases}$$

$$x_{i3} = \begin{cases} 1 & \text{IF ANTIDOTE C} \\ 0 & \text{OTHERWISE} \end{cases}$$

$$\bullet \hat{\beta}_1 = \bar{Y}_A$$

$$\bullet \hat{\beta}_1 + \hat{\beta}_2 = \bar{Y}_B \longrightarrow \hat{\beta}_2 = \bar{Y}_B - \bar{Y}_A$$

$$\bullet \hat{\beta}_1 + \hat{\beta}_3 = \bar{Y}_C \longrightarrow \hat{\beta}_3 = \bar{Y}_C - \bar{Y}_A$$

$$\hat{\beta}_1 = 0.347$$

$$\hat{\beta}_2 = 3.067 - 0.347 = 2.72$$

$$\hat{\beta}_3 = 1.764 - 0.347 = 1.417$$

$$Y_i = 0.347 + 2.72x_{i2} + 1.417x_{i3}$$

$$\bullet \hat{\beta}_1 = 0.347 \longrightarrow \text{AVERAGE OF SURVIVAL TIME (Y) WHEN GROUP IS A}$$

$$\bullet \hat{\beta}_2 = 2.72 \longrightarrow \text{AVERAGE INCREASE IN SURVIVAL TIME Y WHEN GROUP IS B}$$

$$\bullet \hat{\beta}_3 = 1.417 \longrightarrow "$$

" WHEN GROUP IS C

Exercise 3.3

Propose an alternative test for ex. 3.1 specifying the system of hypothesis. Then, compute R^2 of our model.

4 Exercise 4

Lab *Precise* got some measurements to see whether the tar contents (in milligrams) for three different brands of cigarettes are different. The measurements are showed in the following table.

Sample	Brand A	Brand B	Brand C
1	10.21	11.32	11.60
2	10.25	11.20	11.90
3	10.24	11.40	11.80
4	9.80	10.50	12.30
5	9.77	10.68	12.20
6	9.73	10.90	12.20

Write the equation of the linear regression model. Find and interpret the estimates of regression coefficients.

Exercise 4.1

Complete the following table.

Sources of variation	D.o.f.	Deviance
Total		
Regression		
Residual		

Perform the statistical test to evaluate if the means (of each group) are homogeneous.